Meaning–Inertia

An Ontological Substrate Beneath Inertia, Gravitation, and Spacetime

In this framework, inertia and gravitation are not forces but manifestations of a state-dependent participation scalar.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Modern physics successfully predicts phenomena across vast scales, yet leaves several foundational questions structurally unresolved: the origin of inertia, the exact equivalence of inertial and gravitational mass, the ontological status of vacuum energy, and the persistence of de Sitter–like geometry in the absence of matter. This paper proposes a candidate substrate - **Meaning–Inertia (M-I)** - that governs the degree to which physical systems participate in the emergent spacetime projection. In this framework, inertia and gravitation are not forces but manifestations of a state-dependent participation scalar. We articulate the minimal ontological commitments of M-I, identify the constitutional invariants it must obey, state an explicit impossibility that sharpens its identity by excluding classes of counterfactual worlds standard physics leaves formally unconstrained, and enumerate empirical null tests available within existing metrology. No experimental mechanism is proposed; the paper is concerned solely with ontology and constraints, and with conceptual falsifiability rather than immediate detectability.

---

1. Motivation: The Persistent Gap

Despite General Relativity's geometric unification of gravitation and inertia, the coupling itself remains axiomatic. Inertial structure exists even in matter-free solutions; vacuum energy behaves unlike ordinary stress–energy; and nonlocal correlations challenge locality without violating causality. These tensions suggest not missing interactions, but a missing layer of description beneath spacetime dynamics.

Current frameworks either treat inertia as primitive or absorb it into geometry without explaining why geometry resists change in precisely the way it does. The persistence of geometric structure in vacuum solutions, and the stubborn

smallness yet nonzero character of cosmic acceleration, both point toward an ontological substrate that constrains participation in spacetime without itself being spacetime.

Einstein's work provides a methodological template for addressing such gaps. In his distinction between *principle theories* and *constructive theories*, principle theories constrain what is physically admissible without first supplying underlying mechanisms. The equivalence of inertial and gravitational mass was treated by Einstein not as an accidental regularity but as a sign that a deeper unity was required, motivating the geometric formulation of gravitation. Meaning–Inertia adopts a similar posture one layer deeper: it does not propose a new interaction or field equation, but a constitutional substrate that constrains what kinds of continuation and rebinding are admissible at all. In this sense, the proposal is less a competitor to General Relativity than an ontological attempt to explain why the invariances that make GR possible - especially equivalence and the persistence of geometric structure - have the authority they do.

---

2. Ontological Postulate

We postulate a substrate - **Meaning–Inertia (M-I)** - that determines the resistance of a system to rebinding under transformation. Physical quantities ordinarily treated as primitive (inertia, gravitational coupling, admissible motion) arise as projections of M-I state.

Meaning–Inertia is not a medium in which spacetime resides. Rather, spacetime is what counts as realized structure when participation is nonzero. M-I is prior to spacetime **in the sense of constraint and counterfactual dependence**, not in temporal order or causal sequence: altering participation alters the admissible geometry, whereas altering geometry alone does not license unconstrained participation.

2.1 Clarification on the Term "Meaning"

In this work, *meaning* is not used in a semantic, intentional, or observer-relative sense. It denotes the **structural property of a system that determines which transformations are admissible continuations and which incur cost or refusal**. Meaning, in this sense, is binding relevance: the degree to which internal structure constrains lawful continuation.

The term is retained deliberately despite its philosophical weight. No existing physical vocabulary captures constraint relevance without smuggling in dynamics, representation, or agency. Inertia is the physical manifestation of meaning - **resistance to unauthorized change**. Meaning–Inertia therefore names a substrate governing constraint relevance and binding persistence, prior to force, dynamics, or interpretation. This notion of meaning concerns admissibility rather than representation.

The framework treats admissibility, refusal, and binding cost as primitive features of the substrate. Whether this normativity is globally fixed or historically contingent upon accumulated binding constraints is left open at the ontological level; the present work marks this as a boundary rather than a gap.

2.2 Participation Scalar

A system's participation is characterized by a scalar ($\gamma(s)$), dependent on internal state (s), such that

$$m_i(s) = m_g(s) = \gamma(s) \cdot m_0.$$

Here (m_0) denotes the reference mass under full participation. The scalar ($\gamma(s)$) is not freely specifiable: for most ordinary state variations it is constitutionally constrained to unity within experimental bounds. Any admissible variation must respect the invariants articulated below, including global accounting, locality of projection, and thermodynamic honesty.

The framework constrains ($\gamma(s)$) negatively. In particular, (γ) cannot coherently depend on coordinate description, observer-relative coarse-graining, reversible bookkeeping transformations, or internal reparameterizations that leave all binding relations intact. Any apparent variation not accompanied by compensating binding cost elsewhere is inadmissible.

Inertial and gravitational mass are equal because both arise as projections of ($\gamma(s)$), not because of unexplained coincidence.

2.3 An Explicit Impossibility

To sharpen the ontological identity of Meaning–Inertia, we state one impossibility that holds within the framework, even in principle:

Impossibility (I): No transformation that preserves all binding relations of a system - i.e., that is reversible, purely descriptive, or exhaustively reducible to mass–energy bookkeeping - can alter ($\gamma(s)$).

Equivalently, participation cannot change without the creation, destruction, or redistribution of binding constraints. Any purported state change that affects inertia or gravitation without incurring binding cost violates the constitution of M-I and is inadmissible. This impossibility is intended to exclude at least one coherent-seeming class of counterfactual worlds - those in which geometry or inertia rebinds freely under binding-preserving transformations - that standard physics leaves formally unconstrained "in principle."

---

3. Consequences for Spacetime and Cosmology

Within the M-I framework:

1. **Spacetime geometry is emergent, not fundamental.** Geometry reflects collective participation constraints rather than primitive structure.
2. **de Sitter spacetime is reinterpreted as a default projection regime of minimal binding constraint**, not as "empty reality" endowed with a material vacuum energy source.
3. **Cosmic acceleration reflects approach toward an idle participation regime**, rather than literal gravitation of vacuum energy.

The term *approach* denotes a constitutional limit on admissible projection states, not a temporal relaxation process within spacetime. Geometry can therefore exist without localized matter because M-I persists without localized commitments.

Pure geometric autonomy is insufficient to account for this persistence. Geometry alone can encode relations, but not why those relations resist arbitrary rebinding. Treating geometry as self-sufficient leaves unexplained why maximally permissive geometries are not realized. M-I supplies the missing refusal: the constraint that prevents geometry from dissolving into unrestricted continuation. This reinterpretation does not compete with quantitative cosmology; it addresses the ontological status presupposed by those quantities.

---

4. Constitutional Invariants of the M-I Substrate

Meaning–Inertia is not a dynamical theory but a constitutional substrate. For it to be lawful, it must enforce the following invariants:

1. **Global Accounting** Energy–momentum is conserved across system plus substrate.
2. **Substrate Causality** A causal partial order exists at the M-I level; projection cannot create causal loops.
3. **Binding Irreversibility** Established constraints cannot be annulled without compensating rebinding cost.
4. **Locality of Projection** Local participation changes need not induce global geometric back-reaction.
5. **Thermodynamic Honesty** Total entropy is nondecreasing across closed systems. Entropy appears here as a constitutional bookkeeping of irreversible binding change, not merely as an emergent statistical measure.
6. **Refusal Thresholds** Continuation may fail when binding constraints are violated.

These invariants act as a constitution rather than as equations of motion. Any instantiation of M-I that violates them is inadmissible.

---

5. Relation to Existing Frameworks

Newtonian mechanics treats inertia as primitive.

General Relativity geometrizes inertia but leaves the coupling unexplained.

Quantum theory reveals nonlocal structure without positing an ontological substrate beneath spacetime.

Machian intuitions correctly identify inertia as non-primitive but lack formal completion.

Meaning–Inertia supplies the missing substrate without modifying established predictive formalisms. All known laws are recovered as projection-level regularities. Even if $(\gamma(s))$ is found to be exactly constant, explaining **why** universality must hold remains an ontological task distinct from assuming it.

---

6. Empirical Handles and Null Tests (Non-Exhaustive)

Although ontological in scope, the framework is falsifiable. If $(\gamma(s))$ is nontrivial, ordinary physics admits multiple null tests. The diagnostic force of these tests lies as much in persistent invariance as in deviation.

(T1) State-Dependent Universality of Free Fall (WEP/UFF) Prediction: Identical bodies prepared in different internal states may exhibit distinct $(\gamma(s))$, beyond standard mass–energy accounting. Null expectation: $(\gamma(s) = 1)$.

(T2) State-Dependent Gravitational Redshift (LPI) Prediction: The redshift relation may acquire a small state-dependent correction. Null expectation: No dependence beyond known systematics.

(T3) Parametric Inertial Response Prediction: High-Q oscillators may show correlated spectral sidebands under controlled internal state modulation. Null expectation: No sidebands beyond established coupling channels.

(T4) Vacuum Energy Sequestering Prediction: Vacuum energy contributions do not gravitate naively; effective decoupling occurs at the participation level. Null expectation: Vacuum energy couples as ordinary stress–energy.

(T5) Coherence-Dependent Participation Prediction: Identical matter undergoing irreversible binding changes may show differential inertial or gravitational response not reducible to $(E = mc^2)$. Null expectation: All effects reduce to standard accounting.

Persistent null results constrain $(\gamma(s))$; nonzero detections motivate quantitative refinement.

---

7. Scope and Limits

This paper proposes no mechanism, no device, and no control scheme. Its aim is to identify a missing ontological layer and the constraints it must obey. Whether and how Meaning–Inertia may be accessed experimentally is explicitly left open.

---

8. Conclusion

Meaning–Inertia reframes inertia and gravitation as questions of participation rather than force. By introducing a lawful substrate beneath spacetime, it explains why laws have staying power: because not all continuations are permitted. Something binds. Something refuses. By making that refusal explicit, the framework resolves long-standing conceptual tensions while preserving established physics. The proposal stands or falls by ordinary measurements, not by rhetoric.

Appendix A: Falsifiability and Conditions of Failure

A.1 Falsifiability as Used Here

In this work, *falsifiable* means that there exist conceivable observations, expressible in ordinary physics language and accessible to current or near-current metrology, that would contradict the core postulates of the framework.

The framework does not claim immediate detectability, large-magnitude effects, or exclusivity over existing theories. It does claim that if Meaning–Inertia is real, its effects cannot be perfectly hidden across all domains where inertial and gravitational universality are tested under controlled state variation.

A.2 Core Commitments Subject to Test

The framework makes three commitments that are, in principle, testable:

1. **State Dependence** The participation scalar ($\gamma(s)$) may depend on internal state (s), not solely on rest mass and composition.
2. **Structural Equivalence** Inertial and gravitational mass are equal because both arise from ($\gamma(s)$).
3. **Projection-Level Universality** Standard laws of motion and gravitation hold as projection-level rules, not as ontological primitives.

A.3 Conditions That Would Count Against Meaning–Inertia

The framework would be significantly undermined if any of the following were established to arbitrary precision:

(F1) Absolute State-Independence of Inertia and Gravitation If a single physical system, prepared in arbitrarily distinct internal states, exhibits strictly identical inertial and gravitational response beyond standard mass–energy accounting, then $(\gamma(s))$ collapses to a universal constant.

(F2) Exhaustive Reduction to Mass–Energy Equivalence If all observed differences reduce uniquely to $(E = mc^2)$ bookkeeping, Meaning–Inertia provides no additional explanatory power.

(F3) Inconsistency with Conservation or Causality Any violation of global conservation, causal order, or entropy monotonicity renders the ontology inadmissible.

(F4) Naive Gravitational Coupling of Vacuum Energy If vacuum energy is shown to gravitate exactly as ordinary stress–energy without sequestering or reparameterization, the reinterpretation of de Sitter geometry fails.

(F5) Asymptotic Irrelevance If increasingly sensitive searches exclude any nontrivial $(\gamma(s))$ dependence across all accessible regimes, the hypothesis should be regarded as an unnecessary ontological layer.

A.4 What Would Not Falsify the Framework

The following do not constitute falsification:

- Absence of large or easily detectable effects.
- Requirement of extreme precision.
- Strong upper bounds on $(\gamma(s))$.
- Failure of any particular experimental proposal.

Meaning–Inertia is constrained by patterns of impossibility, not by the success of a device.

A.5 Status of the Hypothesis

Meaning–Inertia is a candidate substrate ontology that resolves conceptual tensions, preserves known laws at the projection level, and admits clear conditions under which it would fail. It is neither empirically confirmed nor empirically insulated - a lawful possibility whose relevance must be decided by ordinary measurement rather than rhetorical preference.

Corpus-Wide Reference Positioning

This paper argues that modern physics successfully predicts phenomena across vast scales, yet leaves several foundational questions structurally unresolved: the origin of inertia, the exact equivalence of inertial and gravitational mass, the ontological status of vacuum energy, and the persistence of de Sitter–like geometry in the absence of matter. Its nearest external comparison fields are semantic, consciousness, governance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counter-pressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Structural and computational correspondences remain insufficient as phenomenal evidence unless linked to discriminating empirical predictions against competing theories. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] David J. Chalmers. "Facing Up to the Problem of Consciousness." *Journal of Consciousness Studies* 2(3), 1995, pp. 200-219. https://consc.net/papers/facing.html **Relationship:** Foundational framing of the explanatory gap; cited to bound, not resolve, phenomenal claims.

[**E7**] Larissa Albantakis et al. "Integrated Information Theory (IIT) 4.0: Formulating the Properties of Phenomenal Existence in Physical Terms." *PLOS Computational Biology* 19(10), 2023, e1011465. DOI: 10.1371/journal.pcbi.1011465. https://doi.org/10.1371/journal.pcbi.1011465 **Relationship:** Competing formal theory of consciousness included for explicit contrast and counterevidence obligations.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-358AC09F5A13587E`;
source SHA-256
`AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-091E34BA93992CE4`;
source SHA-256
`DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Ableman Consciousness Ontology (ACO).* Ableman Corpus current edition; work identity `PAPER.PUB.18203749`;
source SHA-256
`ACD97032780DA7B3A36750702BE1F47BCB2E6361EF6E74ECA8B441D7C6A25DDE`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *The Locus of Qualia.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-AF27E66B81811DB1`;
source SHA-256
`7B83EBD0FFDCFEBFA522A715AE0E4D1049FCA0B4CE0618C2017F8BB9206E5424`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.18209065